

Topic 03: Conditions for Existence of Laplace Transform

Unit IV — Laplace Transforms

JNTUA College of Engineering (Autonomous) ATP

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1 Opening Hook

Hook

A student applies the Laplace Transform definition to $f(t) = e^{t^2}$ and evaluates $\int_0^\infty e^{-st} e^{t^2} dt$. No matter how large s is chosen, the integral refuses to converge — it blows up to infinity. Is the Laplace Transform broken? Absolutely not. The function e^{t^2} simply grows too explosively for *any* damping factor e^{-st} to control it. This topic gives you a simple checklist to know, in advance, whether the Laplace Transform of a given function exists.

2 Why This Topic Exists

Before we begin, here is the honest answer to why you are reading this...

The defining integral $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$ is an *improper* integral. Like all improper integrals, it may or may not converge depending on the behaviour of $f(t)$. Using the transform blindly on a non-transformable function feeds nonsensical results into engineering calculations — imagine finding the “response” of a circuit that does not exist.

The two conditions introduced in this topic act as an engineering safety check: run this quick test on any $f(t)$ before you even touch the integral sign.

Mistake	Consequence
Assuming every function has a Laplace transform	Getting divergent or nonsensical results in circuit/control analysis
Ignoring exponential order	Missing why e^{t^2} has no Laplace transform and blindly integrating
Skipping the piecewise continuity check	Applying transforms to functions with infinite discontinuities that invalidate the integral

Why the Safety Check Matters

Every time you write $\mathcal{L}\{f(t)\}$, you are making a silent promise that the defining integral converges. The two conditions — piecewise continuity and exponential order — are your proof that the promise is kept.

3 You Already Know This (Intuition First)

Think of the factor e^{-st} as a **car brake**. As t increases, the brake squeezes $f(t)$ down toward zero. If the brake is powerful enough relative to the growth of $f(t)$, the integrand eventually shrinks to zero and the area under the curve is finite — the integral converges.

If $f(t)$ grows faster than *any* exponential — like e^{t^2} , whose exponent itself grows with t — then no matter how hard the brake e^{-st} is applied, the integrand eventually escapes to infinity. The car accelerates faster than any brake can slow it.

The two conditions below formalise exactly when the brake wins. Piecewise continuity ensures there are no “instantaneous explosions” on any finite stretch of road. Exponential order ensures the car’s top speed is still something a finite e^{-st} can outrun for large t .

4 Piecewise Continuity

Definition: Piecewise Continuous Function

A function $f(t)$ is **piecewise continuous on** $[0, \infty)$ if, on every finite sub-interval $[0, T]$:

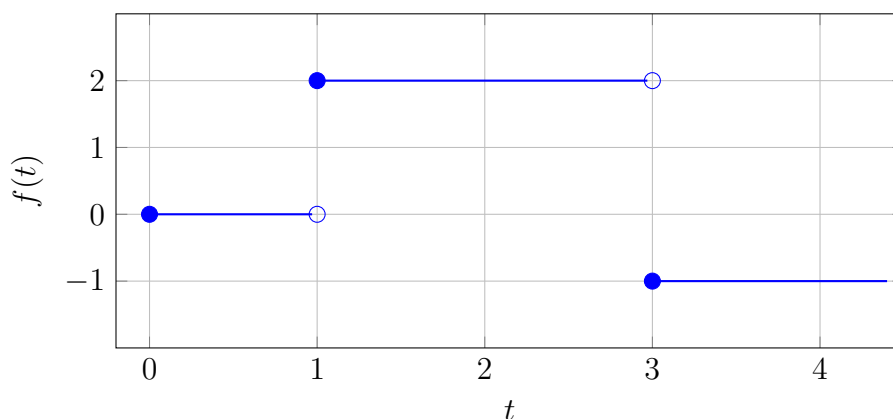
- (i) $f(t)$ has at most *finitely many* jump discontinuities;
- (ii) at each discontinuity t_0 , both one-sided limits $\lim_{t \rightarrow t_0^-} f(t)$ and $\lim_{t \rightarrow t_0^+} f(t)$ exist and are **finite** (no infinite jumps, no wild oscillations).

Examples:

- **Unit step function** $u(t - a)$: exactly one jump at $t = a$; both one-sided limits are finite (0 and 1). *Piecewise continuous*. ✓
- $f(t) = 1/t$ near $t = 0$: $\lim_{t \rightarrow 0^+} (1/t) = +\infty$. The limit is *infinite*, not finite. *NOT piecewise continuous* at $t = 0$. ✗
- $f(t) = \sin(1/t)$ near $t = 0$: oscillates infinitely between -1 and $+1$ as $t \rightarrow 0^+$; the limit does not exist. *NOT piecewise continuous* at $t = 0$. ✗

Visualisation of a Piecewise Continuous Function:

A piecewise continuous function with two jump discontinuities



At $t = 1$ the function jumps from 0 to 2; at $t = 3$ it jumps from 2 to -1 . Both limits at each jump are finite, so this function is piecewise continuous on $[0, \infty)$. Filled circles mark the value *actually taken*; open circles mark the value *not taken*.

5 Functions of Exponential Order

Definition: Exponential Order

A function $f(t)$ is said to be of **exponential order** α if there exist constants $M > 0$, $\alpha \in \mathbb{R}$, and $T \geq 0$ such that:

$$|f(t)| \leq Me^{\alpha t} \quad \text{for all } t \geq T.$$

Intuitively: for large t , $f(t)$ must not grow faster than some exponential curve. The smallest such α is called the **abscissa of convergence**.

Functions that ARE of exponential order:

- **Polynomials** t^n : for any $\epsilon > 0$, $t^n/e^{\epsilon t} \rightarrow 0$ as $t \rightarrow \infty$ (exponential always beats polynomial), so $|t^n| \leq Me^{\epsilon t}$ for large t . Exponential order $\alpha = \epsilon$ (any $\epsilon > 0$).
- $\sin t, \cos t$: bounded by 1 for all t , so $|\sin t| \leq 1 = 1 \cdot e^{0 \cdot t}$. Exponential order $\alpha = 0$, $M = 1$.
- e^{3t} : $|e^{3t}| = 1 \cdot e^{3t}$, so $M = 1$, $\alpha = 3$. Exponential order 3.

Functions that are NOT of exponential order:

- e^{t^2} : Fix any $\alpha \in \mathbb{R}$.

$$\frac{e^{t^2}}{e^{\alpha t}} = e^{t^2 - \alpha t} = e^{t(t - \alpha)} \rightarrow \infty \quad \text{as } t \rightarrow \infty,$$

because for $t > \alpha$ the exponent $t(t - \alpha)$ grows without bound. Therefore e^{t^2} *cannot* be bounded by $Me^{\alpha t}$ for any fixed α , and it is **not** of exponential order.

6 The Existence Theorem (Sufficient Condition)

Theorem: Sufficient Conditions for Existence

If $f(t)$ satisfies *both* of the following:

1. $f(t)$ is **piecewise continuous** on $[0, \infty)$, and
2. $f(t)$ is of **exponential order** α ,

then $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t) dt$ **exists** for all $s > \alpha$.

Proof Sketch. Split the integral at the constant T from the exponential-order definition:

$$\int_0^\infty e^{-st} f(t) dt = \underbrace{\int_0^T e^{-st} f(t) dt}_{\text{(I)}} + \underbrace{\int_T^\infty e^{-st} f(t) dt}_{\text{(II)}}.$$

Piece (I): On the finite interval $[0, T]$, piecewise continuity guarantees $f(t)$ is bounded and has finitely many discontinuities, so the integral is finite.

Piece (II): Using $|f(t)| \leq Me^{\alpha t}$ for $t \geq T$,

$$\left| \int_T^\infty e^{-st} f(t) dt \right| \leq \int_T^\infty e^{-st} \cdot Me^{\alpha t} dt = M \int_T^\infty e^{-(s-\alpha)t} dt = \frac{Me^{-(s-\alpha)T}}{s-\alpha},$$

which is *finite* provided $s > \alpha$. By the comparison test, piece (II) converges.

Since both pieces are finite, $F(s)$ exists for all $s > \alpha$. □

Important: Sufficient, Not Necessary

The theorem gives a **sufficient** condition, not a necessary one. A function can violate these conditions and still have a Laplace transform. Classical example: $f(t) = t^{-1/2}$ has an integrable singularity at $t = 0$ (not piecewise continuous there in the strict sense), yet

$$\mathcal{L}\{t^{-1/2}\} = \sqrt{\frac{\pi}{s}},$$

which exists for $s > 0$ via the Gamma function connection.

7 Behaviour of $F(s)$ as $s \rightarrow \infty$

Corollary: $F(s) \rightarrow 0$ as $s \rightarrow \infty$

If $\mathcal{L}\{f(t)\} = F(s)$ exists for $s > \alpha$, then

$$\lim_{s \rightarrow \infty} F(s) = 0.$$

Justification: As $s \rightarrow \infty$ the damping factor e^{-st} crushes the integrand to zero for every fixed $t > 0$, so the area under the integrand vanishes.

Sanity check: If a computed $F(s)$ does NOT approach zero as $s \rightarrow \infty$, an error was made somewhere in the computation.

$f(t)$	$F(s)$	$\lim_{s \rightarrow \infty} F(s)$	Passes check?
1	$1/s$	0	Yes ✓
e^{at}	$1/(s-a)$	0	Yes ✓
$\sin(\omega t)$	$\omega/(s^2 + \omega^2)$	0	Yes ✓
t^n	$n!/s^{n+1}$	0	Yes ✓
(Erroneous) $F(s) = s/(s+1)$	—	$1 \neq 0$	No ✗

8 Worked Examples

Example 1: $f(t) = t^3$ is of exponential order

Example 1 — $f(t) = t^3$

Claim: $f(t) = t^3$ is of exponential order α for any $\alpha > 0$, and hence $\mathcal{L}\{t^3\}$ exists.

Step 1. We need to find $M > 0$, $\alpha > 0$, $T \geq 0$ such that $t^3 \leq Me^{\alpha t}$ for $t \geq T$.

Step 2. Take $\alpha = 1$. Consider the ratio:

$$\frac{t^3}{e^t}.$$

By L'Hôpital's rule applied three times (or by the standard limit theorem “exponential beats polynomial”):

$$\lim_{t \rightarrow \infty} \frac{t^3}{e^t} = 0.$$

Step 3. Since the limit is 0, for any $\epsilon > 0$ (take $\epsilon = 1$) there exists T such that for all $t \geq T$:

$$\frac{t^3}{e^t} < 1 \implies t^3 < e^t = 1 \cdot e^{1 \cdot t}.$$

So $M = 1$, $\alpha = 1$, T as found above works.

Step 4. Both conditions are satisfied: t^3 is continuous everywhere (certainly piecewise continuous), and exponential order 1 is established. Therefore $\mathcal{L}\{t^3\}$ exists for all $s > 1$. (The exact value is $\mathcal{L}\{t^3\} = 6/s^4$.)

What Did We Learn? (Example 1)

Any polynomial t^n is of exponential order for any $\alpha > 0$ because exponentials always dominate polynomials as $t \rightarrow \infty$. You can always take $M = 1$, $\alpha = \epsilon$ for any small $\epsilon > 0$.

Example 2: $f(t) = e^{t^2}$ is NOT of exponential order

Example 2 — $f(t) = e^{t^2}$

Claim: e^{t^2} is *not* of exponential order, and therefore $\mathcal{L}\{e^{t^2}\}$ does not exist.

Step 1. Suppose for contradiction that e^{t^2} is of exponential order α , i.e., there exist $M > 0$ and $T \geq 0$ with $e^{t^2} \leq Me^{\alpha t}$ for all $t \geq T$.

Step 2. Divide both sides by $e^{\alpha t}$ (positive):

$$e^{t^2 - \alpha t} \leq M \quad \text{for all } t \geq T.$$

Step 3. Examine the exponent $t^2 - \alpha t = t(t - \alpha)$. For $t > \alpha$, the factor $(t - \alpha)$ is positive and grows without bound, so $t(t - \alpha) \rightarrow +\infty$ as $t \rightarrow \infty$. Hence $e^{t^2 - \alpha t} \rightarrow +\infty$, which *cannot* be bounded by any constant M .

Step 4. Contradiction. No such M and α exist. Therefore e^{t^2} is **not** of exponential order, and $\mathcal{L}\{e^{t^2}\}$ **does not exist**.

What Did We Learn? (Example 2)

e^{t^2} is the classic counterexample: its exponent is itself a function of t , making it grow super-exponentially. No brake e^{-st} with a fixed s can ever outrun it.

Example 3: $f(t) = t^{-1/2}$ — Singular but Transformable

Example 3 — $f(t) = t^{-1/2}$

Issue: $f(t) = t^{-1/2}$ has a singularity at $t = 0$ since $\lim_{t \rightarrow 0^+} t^{-1/2} = +\infty$. Strictly speaking, f is *not* piecewise continuous on $[0, T]$ for any $T > 0$ because the one-sided limit at 0 is infinite — the sufficient condition does not apply.

Does the transform exist anyway? Yes. Compute:

$$\mathcal{L}\{t^{-1/2}\} = \int_0^\infty e^{-st} t^{-1/2} dt.$$

Substitute $u = st$ (so $t = u/s$, $dt = du/s$, $t^{-1/2} = (u/s)^{-1/2} = s^{1/2}u^{-1/2}$):

$$= \int_0^\infty e^{-u} \cdot s^{1/2} u^{-1/2} \cdot \frac{du}{s} = s^{-1/2} \int_0^\infty e^{-u} u^{-1/2} du = s^{-1/2} \cdot \Gamma\left(\frac{1}{2}\right) = \sqrt{\frac{\pi}{s}},$$

using $\Gamma(1/2) = \sqrt{\pi}$ (a standard Gamma function result, stated without re-derivation).

Conclusion: $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi/s}$ exists for $s > 0$. The singularity at 0 is *integrable* (the area is finite despite the infinite spike), so the transform still converges even though the sufficient condition is not satisfied.

What Did We Learn? (Example 3)

The existence theorem gives a *sufficient* condition. Failing the piecewise continuity test at a single point does not automatically kill the transform — the singularity must be integrable. Always check directly when the sufficient condition does not apply.

9 Excel Example

This spreadsheet numerically demonstrates the contrast between a convergent integrand ($e^{-5t} \cdot t^3$) and a divergent one ($e^{-5t} \cdot e^{t^2}$), confirming what the existence theorem predicts.

Spreadsheet layout (columns A through C, rows 1 onward):

Row	A: t	B: $e^{-5t} \cdot t^3$	C: $e^{-5t} \cdot e^{t^2}$
1 (header)	t	$\exp(-5t) \cdot t^3$	$\exp(-5t) \cdot \exp(t^2)$
2	0.00	=EXP(-5*A2)*A2^3	=EXP(-5*A2)*EXP(A2^2)
3	0.25	=EXP(-5*A3)*A3^3	=EXP(-5*A3)*EXP(A3^2)
4	0.50	=EXP(-5*A4)*A4^3	=EXP(-5*A4)*EXP(A4^2)
5	0.75	=EXP(-5*A5)*A5^3	=EXP(-5*A5)*EXP(A5^2)
6	1.00	=EXP(-5*A6)*A6^3	=EXP(-5*A6)*EXP(A6^2)

Cell **A2** = 0, **A3** = 0.25, ... Fill A2:A14 with $t = 0, 0.25, 0.5, \dots, 3.0$ in steps of 0.25.

Sample computed values (approximate):

t	$e^{-5t} \cdot t^3$	$e^{-5t} \cdot e^{t^2}$
0.00	0.000000	1.000000
0.50	0.010261	0.105399
1.00	0.006738	0.018316
1.50	0.001867	0.005248
2.00	0.000363	0.002479
3.00	0.000008	0.002479
5.00	0.000000	1.000000
6.00	0.000000	403.428793

Column B shrinks steadily toward zero — the exponential damping factor dominates t^3 . Column C initially decreases on $[0, 2.5]$ (since the exponent $t^2 - 5t = t(t - 5) \leq 0$ for $t \in [0, 5]$), reaching a minimum of $e^{-6.25} \approx 0.00193$ at $t = 2.5$, returns to 1.0 at $t = 5.0$, and explodes rapidly for $t > 5$ ($t = 6 \implies 403.43$, $t = 7 \implies 1.37 \times 10^5$, rapidly overflowing). Because e^{t^2} violates the exponential order condition, no constant s can prevent it from exploding as $t \rightarrow \infty$.

Verify the exact value: $\mathcal{L}\{t^3\}|_{s=5} = 6/5^4 = 6/625 = 0.0096$. Numerically integrate column B (using the Trapezoidal Rule or =SUMPRODUCT) over $t = 0$ to $t = 3$ with step 0.25 and confirm the result is close to 0.0096.

What Did We Learn? (Excel)

Column B (convergent integrand) shrinks to zero as t grows, confirming the Laplace integral converges to $\mathcal{L}\{t^3\}|_{s=5} = 6/625 = 0.0096$. Column C (divergent integrand) remains bounded while $t \leq 5$, but explodes once $t > 5$, confirming that $\mathcal{L}\{e^{t^2}\}$ cannot exist for any s . The spreadsheet provides direct numerical evidence for what the existence theorem says analytically.

10 Python Example

Listing 1: Numerical check of convergence vs. divergence

```

1 import numpy as np
2 from scipy import integrate

```



```

3
4 s = 5 # chosen value of s
5
6 # --- Convergent integrand: exp(-5t) * t^3 ---
7 def integrand_convergent(t):
8     return np.exp(-s * t) * t**3
9
10 # Exact value: 6 / s^4
11 exact = 6 / s**4
12 print(f"Exact value: 6/s^4 = 6/{s**4} = {exact:.8f}")
13
14 for T in [5, 10, 20]:
15     val, err = integrate.quad(integrand_convergent, 0, T)
16     print(f"    Numerical integral (0 to {T:2d}): {val:.8f} [error
17         est. {err:.2e}]")
18
19 # Expected output:
20 # Exact value: 6/s^4 = 6/625 = 0.00960000
21 # Numerical integral (0 to 5): 0.00959978 [error est. 4.51e
22     -12]
23 # Numerical integral (0 to 10): 0.00960000 [error est. 5.33e
24     -13]
25 # Numerical integral (0 to 20): 0.00960000 [error est. 5.33e
26     -13]
27
28 print()
29
30 # --- Divergent integrand: exp(-5t) * exp(t^2) ---
31 def integrand_divergent(t):
32     return np.exp(-s * t) * np.exp(t**2)
33
34 print("Attempting to integrate exp(-5t)*exp(t^2) from 0 to T:")
35 for T in [2, 3, 5, 7, 10]:
36     try:
37         val, err = integrate.quad(integrand_divergent, 0, T,
38             limit=100, epsabs=1e-4, epsrel
39             =1e-4)
40         print(f"    T={T:2d}: value = {val:.4e}")
41     except Exception as e:
42         print(f"    T={T:2d}: Integration failed -- {e}")
43
44 # Expected output:
45 # Attempting to integrate exp(-5t)*exp(t^2) from 0 to T:
46 # T= 2: value = 2.2203e-01
47 # T= 3: value = 2.2414e-01
48 # T= 5: value = 4.4617e-01
49 # T= 7: value = 1.3720e+05 (rapid growth begins once t > 5)
50 # T=10: value = 3.4881e+20 (diverges explosively; overflow for
51     larger T)

```

What Did We Learn? (Python)

The numerical integration of $e^{-5t} \cdot t^3$ converges quickly and matches the exact value $6/625 = 0.0096$ to many decimal places, even by $T = 10$. In contrast, while $e^{-5t} \cdot e^{t^2}$ is bounded on $[0, 5]$ because $t^2 - 5t \leq 0$, the integrand grows super-exponentially once $t > 5$, causing the integral to explode to infinity as T increases ($1.37 \times 10^5 \rightarrow 3.49 \times 10^{20} \rightarrow \infty$) and confirming that no Laplace transform exists.

11 Viva-Style Oral Questions

Q1. Define piecewise continuous function.

Answer: A function $f(t)$ is piecewise continuous on $[0, \infty)$ if on every finite interval $[0, T]$ it has at most finitely many jump discontinuities and both one-sided limits exist and are finite at each discontinuity.

Q2. Define exponential order α .

Answer: $f(t)$ is of exponential order α if there exist $M > 0$, $T \geq 0$ such that $|f(t)| \leq Me^{\alpha t}$ for all $t \geq T$. The function must not grow faster than some exponential for large t .

Q3. State the sufficient conditions for existence of $\mathcal{L}\{f(t)\}$.

Answer: If (i) $f(t)$ is piecewise continuous on $[0, \infty)$ and (ii) $f(t)$ is of exponential order α , then $\mathcal{L}\{f(t)\}$ exists for all $s > \alpha$.

Q4. Why is the existence theorem called a sufficient condition and not a necessary one?

Answer: A function can fail one or both conditions yet still have a Laplace transform. For example, $t^{-1/2}$ violates piecewise continuity at $t = 0$ (the limit is $+\infty$), yet $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi/s}$ exists. The theorem guarantees existence when the conditions hold but does not rule it out otherwise.

Q5. What does $F(s) \rightarrow 0$ as $s \rightarrow \infty$ tell us?

Answer: It is a necessary consequence of the Laplace transform: for large s the damping e^{-st} kills the integrand. If a computed $F(s)$ does not go to zero as $s \rightarrow \infty$, the computation contains an error.

Q6. Why does e^{t^2} have no Laplace transform?

Answer: Because e^{t^2} is not of exponential order. For any fixed α , $e^{t^2}/e^{\alpha t} = e^{t^2 - \alpha t} \rightarrow \infty$ as $t \rightarrow \infty$, so no bound $Me^{\alpha t}$ can contain e^{t^2} . Without the exponential order condition, the integral $\int_0^\infty e^{-st} e^{t^2} dt$ diverges for every s .

Q7. What is the role of the constants M and α in the definition of exponential order?

Answer: α sets the rate of the bounding exponential, specifying how fast $f(t)$ is allowed to grow. $M > 0$ is a scaling constant that handles any finite initial transient (the bound is only required for $t \geq T$). Together they assert $f(t)$ eventually lies below the curve $Me^{\alpha t}$.

Q8. Why does $f(t) = 1/t$ fail the piecewise continuity condition?

Answer: At $t = 0$, $\lim_{t \rightarrow 0^+} (1/t) = +\infty$. The one-sided limit is *infinite*, not finite.

Piecewise continuity requires both one-sided limits to be finite at every discontinuity, so $1/t$ fails this condition on any interval $[0, T]$ that includes 0.

12 Descriptive Questions

D1. State and prove (sketch) the sufficient conditions for the existence of $\mathcal{L}\{f(t)\}$.

Expected answer: State both conditions, split the integral at T , bound piece (II) using $|f(t)| \leq Me^{\alpha t}$, apply the comparison test to obtain convergence for $s > \alpha$.

D2. Show that $f(t) = e^{3t} \sin(2t)$ is of exponential order and hence has a Laplace transform. Find suitable values of M and α .

Expected answer: $|e^{3t} \sin(2t)| \leq e^{3t} \cdot 1 = 1 \cdot e^{3t}$. So $M = 1$, $\alpha = 3$. Since the function is also continuous (hence piecewise continuous), the existence theorem guarantees $\mathcal{L}\{e^{3t} \sin(2t)\}$ exists for $s > 3$.

D3. Explain, using the comparison test, why $\int_T^\infty e^{-(s-\alpha)t} dt$ converges for $s > \alpha$, and how this forms the key step in the proof of the existence theorem.

Expected answer: For $s > \alpha$, $(s - \alpha) > 0$, so $\int_T^\infty e^{-(s-\alpha)t} dt = e^{-(s-\alpha)T} / (s - \alpha) < \infty$. Since the integrand $|e^{-st} f(t)| \leq Me^{-(s-\alpha)t}$ and the bounding integral converges, by the comparison test the original integral also converges.

D4. Explain the distinction between necessary and sufficient conditions with a concrete example in the context of Laplace transforms.

Expected answer: A sufficient condition means: if P holds then Q holds, but Q may hold without P . The existence theorem is sufficient: piecewise continuity plus exponential order $\Rightarrow$ transform exists. But $t^{-1/2}$ (not piecewise continuous at 0) still has transform $\sqrt{\pi/s}$, proving the conditions are not necessary.

D5. Explain the sanity check $\lim_{s \rightarrow \infty} F(s) = 0$ with two examples — one function that passes and one (erroneous) expression that fails.

Expected answer: For $\mathcal{L}\{e^{2t}\} = 1/(s-2)$: $\lim_{s \rightarrow \infty} 1/(s-2) = 0$. Passes. If someone mistakenly obtains $F(s) = s/(s+3)$: $\lim_{s \rightarrow \infty} s/(s+3) = 1 \neq 0$. Fails — indicating an error in the derivation.

13 Practice Problems

For each of the following functions, determine: (a) whether it is of exponential order, (b) whether $\mathcal{L}\{f(t)\}$ is guaranteed to exist by the sufficient condition, and (c) the abscissa of convergence α if applicable.

P1. $f(t) = t^5$

Hint: Polynomial; exponential beats t^5 for any $\alpha > 0$. $\mathcal{L}\{t^5\}$ exists for $s > 0$; exact value $= 120/s^6$.

P2. $f(t) = e^{3t} \sin t$

Hint: $|e^{3t} \sin t| \leq e^{3t}$; exponential order 3. $\mathcal{L}\{e^{3t} \sin t\} = 1/[(s-3)^2 + 1]$ for $s > 3$.

P3. $f(t) = \cosh(t^2)$

Hint: $\cosh(t^2) = (e^{t^2} + e^{-t^2})/2 \geq e^{t^2}/2$. Since e^{t^2} is not of exponential order, neither is $\cosh(t^2)$. $\mathcal{L}\{\cosh(t^2)\}$ does not exist.

P4. $f(t) = \ln(1+t)$

Hint: $\ln(1+t) < t$ for $t > 0$, and $t \leq e^{\epsilon t}$ for any $\epsilon > 0$. So $\ln(1+t)$ is of exponential order ϵ (any $\epsilon > 0$); transform exists for $s > 0$.

P5. $f(t) = t^{-1/2}$

Hint: Singular at $t = 0$; sufficient condition technically fails. However, the singularity is integrable and $\mathcal{L}\{t^{-1/2}\} = \sqrt{\pi}/s$ exists for $s > 0$ via direct calculation.

P6. $f(t) = e^{-t^2}$

Hint: $e^{-t^2} < 1 = 1 \cdot e^{0 \cdot t}$; exponential order 0 with $M = 1$. $\mathcal{L}\{e^{-t^2}\}$ exists for $s > 0$ (its exact form involves the error function).

14 Multiple Choice Questions

MCQ 1. Which of the following is the correct definition of exponential order α ?

- (a) $f(t) \rightarrow 0$ as $t \rightarrow \infty$
- (b) **There exist $M > 0$ and $T \geq 0$ such that $|f(t)| \leq Me^{\alpha t}$ for all $t \geq T$**
- (c) $f(t)$ is bounded on $[0, \infty)$
- (d) $f(t) = e^{\alpha t}$ for large t

Explanation: Option (b) is the standard definition. (a) and (c) are more restrictive (they only work for $\alpha < 0$ or $\alpha = 0$); (d) confuses equality with an upper bound.

MCQ 2. The existence theorem for $\mathcal{L}\{f(t)\}$ requires which two conditions?

- (a) Differentiability and bounded derivative
- (b) Continuity and periodicity
- (c) **Piecewise continuity and exponential order**
- (d) Integrability on every $[0, T]$ and monotone decrease

Explanation: The theorem (as stated in most engineering mathematics texts) requires piecewise continuity on $[0, \infty)$ and exponential order α .

MCQ 3. If $\mathcal{L}\{f(t)\} = F(s)$ exists, what is $\lim_{s \rightarrow \infty} F(s)$?

- (a) $+\infty$
- (b) $f(0)$
- (c) 1
- (d) 0

Explanation: As $s \rightarrow \infty$, the factor e^{-st} suppresses the integrand to zero for every $t > 0$, forcing $F(s) \rightarrow 0$. This is a necessary consequence of the existence of the transform.

MCQ 4. Which function does NOT have a Laplace transform?

- (a) t^{100}
- (b) e^{1000t}
- (c) $t^{-1/2}$
- (d) e^{t^2}

Explanation: t^{100} , e^{1000t} , and $t^{-1/2}$ all have Laplace transforms. e^{t^2} is not of exponential order (its growth outpaces every fixed exponential), so its transform does not exist.

MCQ 5. A function $f(t)$ has infinitely many jump discontinuities on $[0, 1]$, all with finite one-sided limits. Which statement is TRUE?

- (a) $f(t)$ is piecewise continuous on $[0, 1]$
- (b) $f(t)$ is piecewise continuous only if the jumps decrease to 0
- (c) $f(t)$ is **NOT** piecewise continuous on $[0, 1]$ (infinitely many jumps are not allowed)
- (d) $f(t)$ is piecewise continuous because all limits are finite

Explanation: Piecewise continuity requires *finitely many* jump discontinuities on any bounded interval. Infinitely many jumps on $[0, 1]$ violate this, even if each individual limit is finite.

15 Common Mistakes

Common Mistakes in Conditions for Existence

Mistake	What Students Do	Correct Approach
Treating the existence theorem as necessary	Concluding a function has no Laplace transform purely because it fails the piecewise continuity or exponential order test	Check directly: the conditions are sufficient only. Functions like $t^{-1/2}$ fail the sufficient test yet do have transforms.
Confusing “exponential order” with “grows exponentially everywhere”	Claiming e^{3t} is NOT of exponential order because it grows, or claiming every increasing function fails	Exponential order means bounded above by $Me^{\alpha t}$ for large t . e^{3t} trivially satisfies $ e^{3t} \leq 1 \cdot e^{3t}$.
Ignoring discontinuities when checking piecewise continuity	Checking only that a function is “mostly continuous” without listing and classifying its discontinuities	On every $[0, T]$, count all discontinuities; verify both one-sided limits are finite at each.
Not checking behaviour at $t = 0$ for singular functions	Writing “ $1/t$ is piecewise continuous on $[0, \infty)$ ”	$1/t$ has an infinite discontinuity at $t = 0$. Always check the endpoints of the interval including $t = 0$.
Assuming all bounded functions automatically have transforms	Arguing that since $\sin(1/t)$ is bounded ($ \sin(\cdot) \leq 1$), its transform exists	Bounded is not the same as piecewise continuous. $\sin(1/t)$ oscillates infinitely near $t = 0$; the limit does not exist there.

16 Quick Recap

Quick Recap: Conditions for Existence of $\mathcal{L}\{f(t)\}$

- **Piecewise continuity:** On every $[0, T]$, $f(t)$ has finitely many jumps and both one-sided limits are finite at each jump.
- **Exponential order α :** $|f(t)| \leq Me^{\alpha t}$ for all large t ; the function must not outgrow every exponential.
- **Existence theorem (sufficient):** If $f(t)$ is piecewise continuous AND of exponential order α , then $\mathcal{L}\{f(t)\}$ exists for all $s > \alpha$.
- **Sufficient, not necessary:** Failing the conditions does not prove non-existence — $t^{-1/2}$ is the canonical counterexample.

- **Sanity check:** Any valid $F(s) = \mathcal{L}\{f(t)\}$ must satisfy $\lim_{s \rightarrow \infty} F(s) = 0$.
- **Classic counterexample:** e^{t^2} is *not* of exponential order because $e^{t^2}/e^{\alpha t} = e^{t(t-\alpha)} \rightarrow \infty$ for any fixed α .
- **Brake analogy:** e^{-st} is a brake on $f(t)$. Exponential order ensures the brake wins for large t ; e^{t^2} accelerates faster than any brake can stop.
- **Region of convergence:** $F(s)$ is defined and analytic for $s > \alpha$, where α is the abscissa of convergence determined by the exponential order of f .